RAILWAY FARE AND RESERVATION CALCULATOR

1. Problem Statement

Develop a railway fare and reservation system that calculates ticket fares for multiple passengers. The system considers travel distance slabs, travel class, age-based concessions, reservation fees, and cancellation penalties. It displays each passenger's fare and the total amount payable or refundable.

2. Objective

To calculate railway ticket charges for multiple passengers by applying distance-based fares, class rates, eligible age concessions, reservation fees, and cancellation penalties.

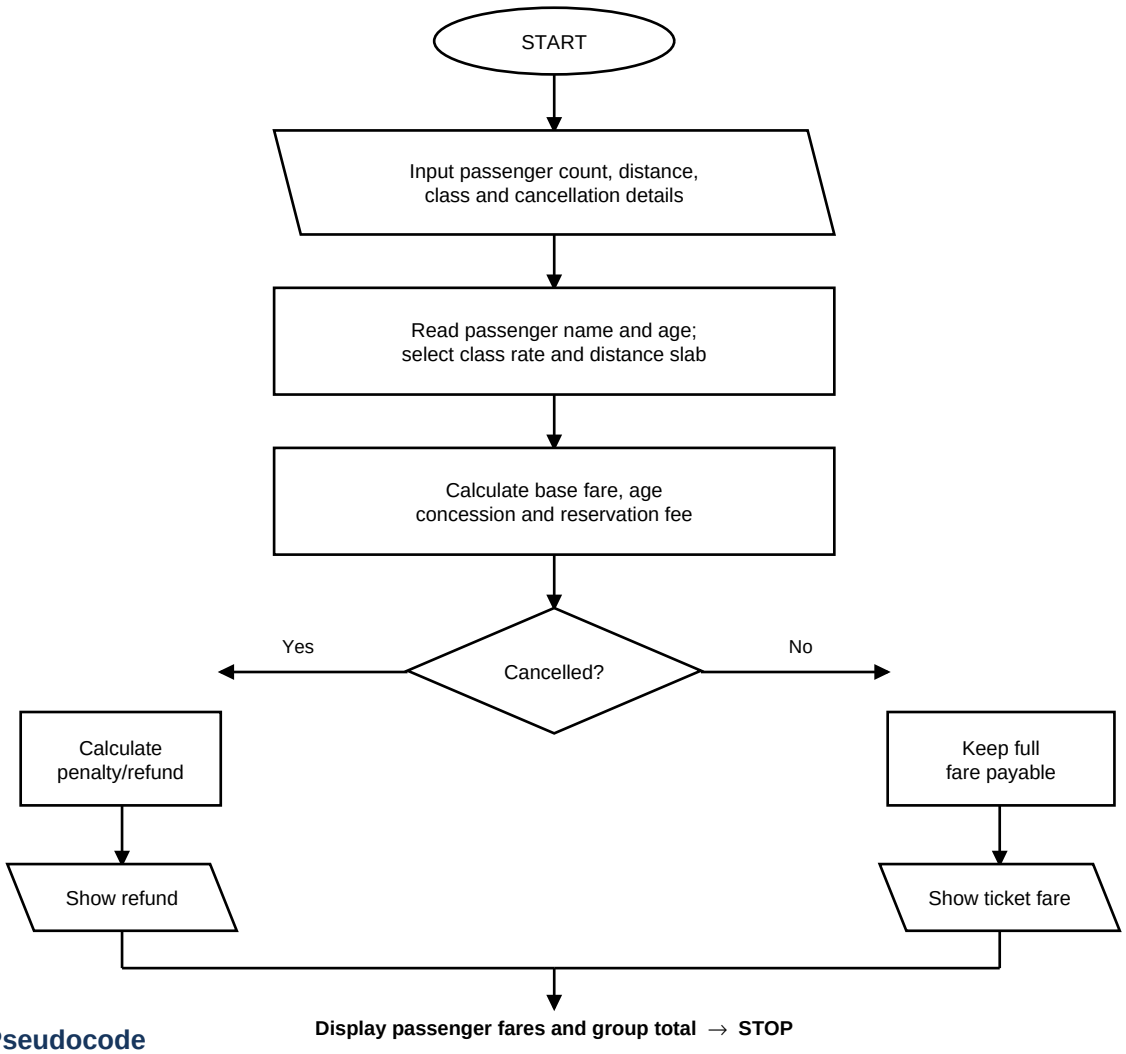
3. Assumptions

1. Fare rates and distance slabs are sample values used for this program.
2. Travel classes are Sleeper, AC 3 Tier, and AC 2 Tier.
3. Children below 5 years travel free; passengers aged 60 or above receive a 20% concession on base fare.
4. Reservation fees are added after applying the age concession.
5. Cancellation penalty is a specified percentage of the ticket total and cannot exceed the ticket total.

4. Algorithm

1. Start.
2. Read passenger count, travel distance, class, and cancellation choice.
3. For each passenger, read name and age.
4. Select the fare rate according to travel class.
5. Find the distance slab and calculate base fare.
6. Apply the eligible age concession to the base fare.
7. Add reservation fee to calculate ticket total.
8. If cancelled, calculate penalty and refundable amount.
9. Display each passenger's fare and group total.
10. Stop.

5. Flowchart (Normal Symbols)



6. Pseudocode

START
INPUT passenger_count, distance, travel_class
INPUT cancellation_choice
SELECT rate based on travel_class
IF distance <= 100 THEN slab_rate = 1.0
ELSE IF distance <= 300 THEN slab_rate = 0.9
ELSE slab_rate = 0.8
group_total = 0
FOR each passenger
 INPUT name, age
 base_fare = distance * rate * slab_rate
 IF age < 5 THEN base_fare = 0
 ELSE IF age >= 60 THEN base_fare = base_fare * 0.80
 ticket_total = base_fare + reservation_fee
 IF cancelled THEN
  penalty = ticket_total * penalty_rate / 100
  refund = ticket_total - penalty
 ELSE
  refund = 0
 END IF
 DISPLAY passenger fare and refund
 group_total = group_total + ticket_total
END FOR
DISPLAY group_total
STOP

7. Python Code

```
print("RAILWAY FARE AND RESERVATION CALCULATOR")

passenger_count = int(input("Enter number of passengers: "))
distance = float(input("Enter travel distance in km: "))
travel_class = input("Enter class (Sleeper/AC3/AC2): ").lower()

if travel_class == "sleeper":
    rate = 1.0
    reservation_fee = 20
elif travel_class == "ac3":
    rate = 2.0
    reservation_fee = 40
elif travel_class == "ac2":
    rate = 3.0
    reservation_fee = 50
else:
    rate = 0
    reservation_fee = 0
    print("Invalid class")

if distance <= 100:
    slab_rate = 1.0
elif distance <= 300:
    slab_rate = 0.9
else:
    slab_rate = 0.8

penalty_rate = float(input("Enter cancellation penalty percentage: "))
cancelled = input("Cancel tickets? (yes/no): ").lower()

group_total = 0
total_refund = 0

for i in range(passenger_count):
    print("Passenger", i + 1)
    name = input("Enter name: ")
    age = int(input("Enter age: "))
    base_fare = distance * rate * slab_rate

    if age < 5:
        base_fare = 0
    elif age >= 60:
        base_fare = base_fare * 0.80

    ticket_total = base_fare + reservation_fee
    group_total = group_total + ticket_total

    if cancelled == "yes":
        penalty = ticket_total * penalty_rate / 100
        refund = ticket_total - penalty
        total_refund = total_refund + refund
    else:
        penalty = 0
        refund = 0

    print("Passenger:", name)
    print("Ticket fare: Rs.", round(ticket_total, 2))
    if cancelled == "yes":
```

```
        print("Cancellation penalty: Rs.", round(penalty, 2))
        print("Refund amount: Rs.", round(refund, 2))

print("Group ticket total: Rs.", round(group_total, 2))
if cancelled == "yes":
    print("Total refund: Rs.", round(total_refund, 2))
```

8. Expected Output

RAILWAY FARE AND RESERVATION CALCULATOR
Enter number of passengers: 2
Enter travel distance in km: 200
Enter class (Sleeper/AC3/AC2): sleeper
Enter cancellation penalty percentage: 10
Cancel tickets? (yes/no): no
Passenger 1
Enter name: Ravi
Enter age: 25
Passenger: Ravi
Ticket fare: Rs. 200.0
Passenger 2
Enter name: Sita
Enter age: 65
Passenger: Sita
Ticket fare: Rs. 164.0
Group ticket total: Rs. 364.0